Reflection Log

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Credit Name: CSE3910 - Project D
Assignment Name: Box

How has your program changed from planning to coding to now?

The objective was to make the rover trace a square path. While the basic logic worked, challenges arose that required fine-tuning for better accuracy.

1. **Inconsistent Turns**
   The rover sometimes over-rotated during turns. I recalibrated the timing for 90-degree turns to match real-world performance.
   Code Example:

```
leftMotors.setTargetVelocity(-1);
rightMotors.setTargetVelocity(1);
Thread.sleep(500); // Adjusted timing for precise turns
```

2. **Uneven Travel Distance**
   Each side of the square wasn't consistent in length due to speed variations. I ensured both motors ran at the same velocity for uniform movement.
   Code Example:

```
leftMotors.setTargetVelocity(1);
rightMotors.setTargetVelocity(1); // Ensured synchronized speeds
```

3. **Environmental Interference**
   Dust and uneven ground affected the rover's performance. I cleaned the surface and added slight speed adjustments to compensate.
   Code Example:

```
leftMotors.setTargetVelocity(0.9);
rightMotors.setTargetVelocity(0.9); // Reduced speed for control
```

4. **Loop Simplification**
   Initially, I wrote separate commands for each side of the square. I later replaced them with a loop for cleaner and more scalable code.

Code Example:

```
for (int i = 0; i < 4; i++) {
    // Move forward
    leftMotors.setTargetVelocity(1);
    rightMotors.setTargetVelocity(1);
    Thread.sleep(1500);

    // Turn 90 degrees
    leftMotors.setTargetVelocity(-1);
    rightMotors.setTargetVelocity(1);
    Thread.sleep(500);
}
```

Final Thoughts: These changes made the program more efficient and reliable. The rover now traces a perfect square consistently, demonstrating precise control and adaptability.